ASSET-LIGHT DELIVERY NETWORK FOR NIGERIA

Product Requirements Document (PRD), Business Plan and Competitive Positioning

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Core thesis: The business is asset-light, but the software and operations must be control-heavy. The company does not need to own vans or shops at launch, but it must own the booking system, custody ledger, route rules, settlement engine, fraud controls and customer trust layer.

Area	Recommendation
Business model	Hybrid pickup/drop-off parcel network using partner shops, partner vans, scheduled urban sweeps and scheduled intercity routes.
First market	Lagos intra-city first, Abuja second, then Lagos-Abuja scheduled intercity corridor.
Customer focus	SME merchants, Instagram/WhatsApp sellers, individual senders, offices and small business bulk senders.
Key advantage	Lower capital requirement plus stronger custody, tracking, fraud control and offline-first operations than informal waybill systems.
MVP rule	Prove scan discipline and route economics before expanding geography.

1. Executive Summary

The proposed business is a technology-enabled logistics network for moving parcels and small goods across Nigerian cities without buying a large fleet at launch. It combines accredited neighbourhood shops, partner-owned vans, modest city cross-docks, scheduled urban sweeps and scheduled intercity trunk routes.

The model is not a pure on-demand dispatch app and it is not a simple motor-park waybill clone. It formalises the familiar Nigerian waybill habit by adding tracking, identity checks, parcel scanning, proof of custody, route validation, OTP pickup and clear claims handling.

The strongest opportunity is to serve social-commerce and SME sellers who need reliable delivery but cannot afford enterprise logistics complexity. The recommended launch sequence is Lagos first, Abuja second, then the Lagos-Abuja scheduled corridor once hub scanning and route discipline are stable.

2. The Problem

Delivery is a major pain point for Nigerian businesses and individuals. Many sellers have products ready to sell, but getting those products safely to buyers is still difficult. They often rely on bus parks, informal drivers, dispatch riders, or manual WhatsApp coordination.

- Parcels can be lost, delayed, damaged or sent to the wrong location.
- Senders and receivers often lack live tracking or proof of handover.
- Informal waybill systems are familiar and useful, but accountability is weak.
- Doorstep delivery can be expensive because of traffic, failed delivery attempts and poor address quality.
- Small sellers lack a simple, trusted, affordable logistics layer for local and intercity delivery.
- Logistics operators face cash-flow risk when partners need fast payment but customers pay late.

3. Market Opportunity

Nigeria already has strong delivery demand. The opportunity is not to create the market from zero, but to organise an existing movement of goods into a more trusted, visible and scalable network.

- Social commerce is growing through WhatsApp, Instagram, TikTok and Facebook sellers.
- SMEs need delivery for customer orders, returns and stock movement.
- Individuals need affordable intercity sending for clothes, documents, household items and small goods.
- Formal courier operators prove that demand exists, but many small users still fall back to informal channels.
- A hub-and-spoke pickup network can reduce failed delivery cost and improve route batching.

4. The Solution

The solution is an asset-light parcel delivery operating system for Nigeria. Users can book a shipment online, through WhatsApp-assisted support, or at a hub counter. The parcel is dropped at a local hub or collected as a premium add-on, scanned, grouped into a route or manifest, moved by partner vans, and released to the receiver using OTP or verified identity.

Layer	Description

Customer layer	Sender web/app, WhatsApp-assisted booking, business portal and receiver tracking page.
Hub layer	Accredited shops act as drop-off, pickup and failed-delivery points.
Sorting layer	Small leased cross-docks sort parcels by route, city, hub and manifest.
Transport layer	Partner-owned vans run scheduled intra-city sweeps and intercity trunk routes.
Control layer	Software manages scanning, custody, pricing, routing, exceptions, settlement and audit logs.

5. Product and Service Lines

Service	Description	Pricing Position
Hub-to-hub economy	Sender drops at origin hub; receiver collects at destination hub.	Cheapest
Door-to-hub	Platform collects from sender; receiver collects at hub.	Medium
Hub-to-door	Sender drops at hub; final delivery goes to receiver address.	Medium / premium
Door-to-door premium	Collected from sender and delivered to receiver address.	Premium
Intercity scheduled	Goods move between city hubs through scheduled trunk routes.	Route and weight based
Business bulk shipment	Many parcels or cartons move together under a controlled manifest.	Bulk discount with controls

6. Target Customers

Customer Segment	Needs	Platform Response
Individual senders	Send personal goods, documents and small parcels.	Simple booking, payment, tracking and OTP collection.
Social-commerce sellers	Deliver customer orders without owning logistics.	Business account, hub drop-off, wallet, bulk upload and WhatsApp/SMS updates.
SMEs and shops	Move stock, fulfil orders and manage returns.	Bulk shipment, reporting, settlement records and service-level options.
Receivers	Know when goods arrive and collect safely.	Tracking page, notifications, OTP and complaint channel.
Hub partners	Earn side income from parcel handling.	Hub PWA, scan-in/out, inventory view and handling fees.

Van owners/drivers	Earn from assigned routes with clear payment rules.	Route app, manifest scan, proof events and settlement view.

7. Why Us: Competitive Differentiation

The company is different because it combines the convenience of Nigerian informal waybill systems with the control discipline of a modern logistics operating system.

Competitor Type	Typical Weakness	Our Difference
Traditional courier firms	Higher infrastructure cost and less flexible for informal SME senders.	Asset-light hubs and partner vans reduce capital intensity while keeping control through software.
Bike dispatch apps	Good for instant local jobs but weak for batching, hub networks and intercity movement.	Scheduled sweeps and intercity trunking improve route economics.
Motor-park waybill	Familiar and flexible, but weak tracking, custody proof and claims handling.	Same convenience with QR/waybill tracking, OTP, scan records and dispute workflow.
Marketplace logistics	Often tied to one platform or seller ecosystem.	Open network for individuals, SMEs, social-commerce sellers and future merchant APIs.
Pure crowdshipping	Flexible but inconsistent for service-level commitments.	Crowd capacity is used only as controlled overflow, not the core promise.
Freight marketplaces	Often focus on trucks, large cargo and enterprise freight.	Parcel and SME focus with hubs, manifests and customer-facing tracking.

Differentiator	Meaning
Asset-light but control-heavy	We do not need to own vans or shops, but we own the software, rules, custody ledger and settlement controls.
Offline-first by design	Hub and driver scans can work during poor internet and sync later with validation.
Neighbourhood pickup network	Local shops become parcel access points, reducing failed doorstep delivery cost.
Custody-first software	Every parcel has a current custodian, scan history, handover record and exception status.
SME and social-commerce focus	Built for WhatsApp sellers, Instagram sellers, boutiques, offices and small business bulk movement.
Cash-flow discipline	Prepaid wallets and proof-triggered settlement reduce the working-capital trap.

8. Product Requirements Document

8.1 Product Goal

Build an MVP that allows senders to create shipments, pay, drop parcels at hubs, track movement, and allow hub agents, drivers and admins to scan, route, verify and settle parcels safely.

8.2 MVP Must-Have Features

- Sender booking for individuals and businesses.
- Weight, size, route and service-type pricing.
- Hub selection and destination validation.
- Waybill number and QR/barcode generation.
- Hub PWA for scan-in, scan-out and receiver release.
- Driver route assignment and route scan confirmation.
- Bulk manifest with seal number, expected count and expected weight.
- Payment gateway and prepaid wallet support.
- SMS/WhatsApp notifications and receiver OTP.
- Admin control tower for live parcels, routes, exceptions, disputes and settlements.

8.3 MVP Should Avoid

- Nationwide launch before scan discipline is proven.
- Cold-chain and pharmaceutical delivery before compliance readiness.
- Large owned warehouses.
- Full crowdshipping as the core service promise.
- Cash-on-delivery at scale in the early stage.
- Complex AI route optimisation before reliable operational data exists.

9. Core Workflows

9.1 Personal Hub-to-Hub Workflow

1. Sender creates shipment through app, web, WhatsApp-assisted form or hub counter.
2. System captures receiver details, destination state, city, local area and preferred hub.
3. System calculates price by weight, size, route and service type.
4. Sender pays or uses wallet balance.
5. System creates waybill number and QR/barcode label.
6. Sender drops parcel at origin hub.
7. Hub agent verifies sender details, checks prohibited-item rules and scans parcel in.
8. Parcel joins a route or manifest after destination validation.
9. Driver collects parcel under two-sided handover.
10. Parcel moves through sorting hub and destination hub.
11. Receiver gets OTP and collects parcel.
12. Hub confirms collection and shipment is completed.

9.2 Business Bulk Workflow

13. Business creates shipments individually, by CSV upload or later through merchant API.
14. System validates addresses and groups parcels by destination corridor.
15. Each parcel receives its own waybill number.

16. Parcels going to the same corridor are added to a bulk manifest.
17. Cross-dock records manifest ID, seal number, expected parcel count and expected weight.
18. Partner van collects the sealed manifest under two-sided handover.
19. Destination hub checks seal, weight, count and individual parcel reconciliation.
20. Exceptions are created automatically for missing, wrong-hub, broken-seal or weight-mismatch cases.

9.3 Failed Home Delivery Workflow

21. Driver attempts doorstep delivery.
22. If receiver is unavailable, address is wrong, phone is unreachable or access is denied, driver records failed delivery reason.
23. Driver uploads GPS, timestamp, note and photo/call evidence where appropriate.
24. Parcel is returned to nearest approved local hub.
25. Receiver receives SMS/WhatsApp notification with hub address and OTP.
26. Receiver collects from local hub and hub completes delivery record.

10. Operating Model

Actor	Primary Responsibility	Key Metrics
Sender	Accurate declaration, packaging, payment and prohibited-item compliance.	Prepayment rate, declaration accuracy, complaint rate.
Hub shop	Parcel acceptance, storage, scan-in/out, receiver verification.	Scan compliance, storage accuracy, queue time, incident rate.
Van owner	Vehicle availability, route completion and safe handling.	On-time route completion, damage claims, route adherence.
Driver	Pickup/drop discipline, live status updates and proof of handoff.	Missed scans, dwell time, failed delivery evidence.
Cross-dock supervisor	Sort accuracy, bag sealing, manifest control and exceptions.	Misroutes, bag integrity, reconciliation accuracy.
Admin team	Approvals, pricing, disputes, settlements and performance control.	Loss rate, complaint closure time, margin by route.

11. Pricing and Unit Economics Logic

Pricing must reflect route cost, handling cost, risk, technology cost, payment cost and contribution margin. The model should not promise the cheapest price for every parcel from day one. It should compete on trust, visibility, convenience and reliability while using hub-to-hub economy pricing for cost-sensitive customers.

Final Price = Base Route Fee + Hub Handling Fee + Actual or Volumetric Weight Fee + Doorstep Pickup Fee + Doorstep Delivery Fee + Insurance/Claims Reserve + Payment Processing Fee + Storage Fee - Bulk or Wallet Discount

Input	Purpose

Actual weight	Charges heavy parcels fairly.
Volumetric weight	Charges large but light parcels fairly.
Corridor	Reflects Lagos-Lagos, Lagos-Abuja or other route cost.
Service type	Differentiates hub-to-hub, hub-to-door, door-to-door and intercity.
Declared value	Supports insurance, liability cap and high-value controls.
Bulk volume	Supports business discount while keeping parcel traceability.
Wallet/prepayment	Rewards prepaid usage and reduces cash-flow risk.

12. Business Model

Revenue Stream	Description
Shipment fees	Main revenue from hub-to-hub, hub-to-door, door-to-hub and intercity delivery.
Premium convenience fees	Doorstep pickup, doorstep delivery, priority delivery and special handling.
Business accounts	Wallets, bulk shipment features, business dashboards and future API access.
Storage fees	Charged when receivers fail to collect after the free storage window.
Insurance/declared value add-on	Optional protection for high-value goods above the standard liability cap.
Future services	Fulfilment support, merchant analytics, marketplace integrations and embedded finance.

13. Technology Architecture

The technology should start as a modular monolith for speed and simplicity, with clean module boundaries. It can later split into services when parcel volume, engineering team size and operational complexity justify it.

Layer	Recommended Stack
Backend	NestJS with TypeScript or Django; modular monolith first.
Database	PostgreSQL with PostGIS for hubs, zones, locations, routes and service areas.
Queue	Redis/BullMQ for sync jobs, notifications, settlement jobs and retries.
Storage	S3-compatible object storage for KYC, parcel photos, seal photos and proof-of-delivery.
Hub app	Progressive Web App for small shops and browser-based access.

Driver app	React Native or Flutter with offline-first scan queue.
Admin dashboard	Next.js control tower for routes, parcels, exceptions, claims and settlements.
Payments	Paystack, Flutterwave, bank transfer and prepaid wallet.
Notifications	SMS, WhatsApp Business API, email and push notification.
Maps	OpenStreetMap/OSRM or GraphHopper; commercial map API where required.

14. Trust, Fraud and Compliance Controls

Control	Rule	Risk Reduced
Two-sided handover	Custody changes only when both sides confirm.	False transfer claims.
Device binding	Hub and driver scans must come from approved devices.	Fake remote scanning.
GPS validation	Scans must match registered hub or route geography.	Wrong-location scans.
OTP release	Receiver collects using OTP or approved override.	Wrong receiver collection.
Seal number	Bulk bags/cartons use unique seals checked at destination.	Tampering and missing parcels.
Weight reconciliation	Expected and received manifest weights are compared.	Hidden removal or substitution.
Settlement hold	Partner payout waits for required scan/proof events.	Paying before completion.
Audit log	All admin overrides are logged with reason and evidence.	Insider manipulation.

15. Pilot and Rollout Plan

Phase	Scope	Goal
Phase 1: Lagos	16 Lagos hubs, Lagos cross-dock, Lagos urban sweeps.	Prove hub scanning, customer demand and route batching.
Phase 2: Abuja	8 Abuja hubs, Abuja cross-dock, district-to-district route sweeps.	Validate second-city operations and partner onboarding.
Phase 3: Lagos-Abuja	Scheduled intercity trunk movement with manifests and cut-off times.	Prove intercity reliability, load factor and reconciliation.
Phase 4: Port Harcourt/Kano prep	Prepare hub mapping, partner onboarding and route economics.	Expand only after KPI thresholds are stable.

Pilot KPI	Target
Scan compliance	95%+ before aggressive geographic expansion.
Loss/damage rate	Below 0.5% in pilot operations.
Complaint closure time	Within 30 days, with clear evidence review.
Parcels per van-day	Improve steadily toward route-level profitability.
Blended parcels/day	Target 500+ parcels/day by month 12 for stronger fixed-cost coverage.
Route gross margin	Measured by route, service type and hub cluster.

16. Key Risks and Mitigation

Risk	Why It Matters	Mitigation
Theft or shrinkage at hubs	Partner shops are not natural warehouses.	Sealed bags, daily reconciliation, CCTV where possible, audits and parcel limits.
Misdeclared parcels	Illegal or prohibited items can create legal and safety risk.	Inspection rights, prohibited-item list, sender declaration and photo-at-acceptance for selected goods.
Weak scan discipline	Without scans the trust layer collapses.	Training, device binding, scan KPIs, settlement holds and supervisor review.
Working-capital squeeze	Fast partner payouts and slow customer receipts can damage cash flow.	Prepaid wallets, limited credit, settlement after verified proof and strict merchant credit rules.
Route under-utilisation	Low parcel volume makes van routes unprofitable.	Cluster hubs, focus on dense corridors and delay intercity expansion until load factor supports it.
Regulatory and compliance exposure	Courier, insurance, vehicle and data rules matter.	Budget for proper licensing, insurance, partner KYC and compliance documentation.

17. Funding and Resource Requirement

A serious 12-month pilot should budget for software build, operations, hub setup, incentives, compliance, insurance, marketing and contingency. The earlier research estimates a lean pilot range of about NGN 100 million to NGN 140 million, depending on licensing, staffing, incentives, facility cost and route density.

Budget Line	Indicative Range
Technology build, integrations, devices and support	NGN 15m - NGN 25m
Core operating team for 12 months	NGN 35m - NGN 50m
Hub setup, signage, training and launch materials	NGN 5m - NGN 10m
Hub and van activation incentives/guarantees	NGN 15m - NGN 25m

Marketing and merchant acquisition	NGN 10m - NGN 18m
Compliance, insurance, legal and audits	NGN 8m - NGN 12m
Contingency	NGN 12m - NGN 20m

18. Implementation Roadmap

27. Register company, confirm licensing route and complete legal/compliance review.
28. Build MVP database schema, role-based authentication and shipment service.
29. Build pricing quote, waybill generation and payment/wallet flow.
30. Build hub PWA for scan-in, scan-out and receiver OTP release.
31. Build driver route app with route stops, manifest scans and issue reporting.
32. Build custody event ledger, destination validation and exception queue.
33. Build bulk manifest, seal and reconciliation logic.
34. Integrate SMS/WhatsApp notification and payment webhooks.
35. Build admin control tower for parcels, hubs, routes, disputes and settlements.
36. Recruit pilot hubs and van owners in Lagos; run controlled soft launch.
37. Optimise route density and scan compliance before adding Abuja.
38. Launch Lagos-Abuja scheduled trunk after local city scan discipline is stable.

19. Final Positioning Statement

The company should position itself as a distributed commerce logistics infrastructure platform for Nigeria. It is not simply a courier app. It is a trusted operating layer that helps people and businesses move goods through local hubs, partner vans and controlled software workflows.

The moat is not ownership of vans. The moat is network density, hub relationships, scan discipline, custody data, settlement controls, merchant trust and offline-first logistics software.

20. Source Notes

This document was prepared from the existing project research and architecture documents supplied in the conversation: Asset-light delivery network for Nigeria; Delivery Solutions in Nigeria; and Asset-Light Delivery Network for Nigeria - Dedicated Software and Backend Architecture Design. Figures and implementation details should be verified with current regulatory, insurance and commercial market quotes before fundraising or launch.